



**INTERNATIONAL
UNMANNED AERIAL VEHICLES
COMPETITION ROTARY WING**

INTERNATIONAL UNMANNED AERIAL VEHICLES COMPETITION

TEAM NAME	MACH-X
TEAM TYPE:	UNIVERSITY TEAM
VEHICLE TYPE:	ROTARY WING
UNIVERSITY NAME:	Pakistan Institute of Engineering and Applied Sciences
COUNTRY:	Pakistan
TEAM RESPONSIBLE:	Ahmed Mustafa

1. ORGANIZATION SUMMARY

The team's organization and the targeted plan for design are presented in this section, which follows as

1.1. Team Organization

The team comprises seven undergraduate students and a team advisor from "The Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad." Each Team member has a sub-team and a specific role assigned to each of them. The task distribution and expertise are given in Table 1, and the sub-team of specific team members is shown in Table 2.

Table 1 Task Distribution

Members	Task distribution and expertise
Advisor	Specializing in autonomous navigation and UAV-assisted communication networks, Research focuses on AI-driven UAV control.
Captain	Autonomous UAV configuration. Subsystem integrations.
Member 1	AI image processing algorithm design & implementation.
Member 2	UAV design, performance parameters, PCB designing
Member 3	3D printing and Mission mechanism design & implementation
Member 4	Hardware development (RF Radio design).
Member 5	Target recognition & information extraction and AI image processing algorithm design
Member 6	AI image processing algorithm design, Payload mechanism design

Table 2 Sub-teams Organization

Teams	Members
UAV Configuration & Performance Parameters	Captain, Member 1, Member 2
Mechanical System, UAV Parameters	Captain, Member 3, Member 4
Mission Mechanism System	Member 3, member 1
Electrical and Electronics hardware development	Member 6, Member 5, member 4
Software and Control System	Member 1, Member 5, Member 2

1.2 Timeline and Development Process

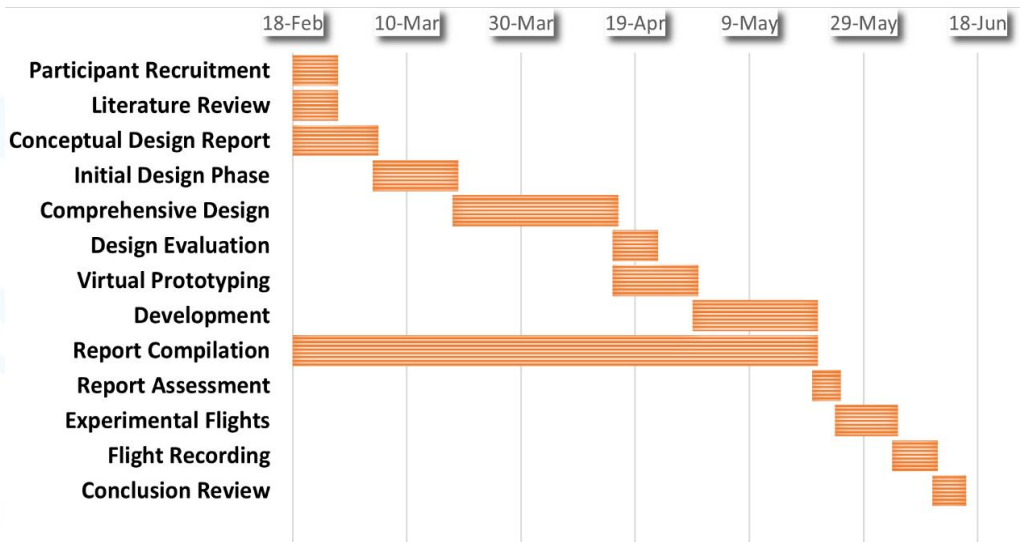


Figure 1 Timeline of project development

2. FEATURES OF UNMANNED AERIAL VEHICLE

2.1 UAV Configuration and Innovative Design Feature

2.1.1 Optimization parameter

The **rotary wing UAV** includes aerodynamic improvements, such as a carbon fiber lightweight and durable frame, landing gear, and vibration dampers, to decrease weight while maintaining strength and ensure precise image processing for target identification. Its squared, ventilated arms increase structural efficiency and enable precise maneuvers, such as circular turns, for the **mechanism of Mission 1**. The payload drop mechanism for the **mechanism of Mission 2** is entirely autonomous. It detects and drops payloads on targets using a Raspberry Pi-based system with machine-learning algorithms for real-time target recognition. The mechanism of mission 3 includes the pick and drop of payloads. UAV will pick up and drop the payload autonomously.

2.1.2 Operational constraint

The UAV is designed to operate within specific boundary conditions to meet optimal conditions and safety. It has a maximum flight time of **6 minutes** with approximately 800g of payload capacity, making it suitable for missions. The maximum weight is limited to 2.5 kg during takeoff to maintain stable flight performance and maneuverability. A remote control (RC) range of up to 1 kilometer (line of sight) provides sufficient operational distance for various emergency conditions. Advanced safety features include **failsafe** modes such as Return-to-Launch (**RTL**) and forced landing, triggered by low battery levels or boundary violations. Additionally, a current breaker interrupts the circuit if battery discharge exceeds 100A, preventing overload or short-circuit conditions, while a manual cut-off circuit allows operators to shut down the system instantly if needed.

Table 3 Suitable design selection

Mechanical Component	Options	Manufacturability (10)	Weight (10)	Stability (10)	Cumulative Assessment (30)
Frame Type	X-Frame	9	9	10	28
	H-Frame	8	8	9	25
	(+) Frame	7	7	8	22
Frame Material	Carbon Fiber	8	10	9	27
	Aluminum	7	7	8	22
	Plastic	9	6	5	20
Base Plate	Carbon Fiber	8	10	9	27
	Aluminum	7	7	8	22
Propellers	Carbon Fiber	9	10	9	28
	Plastic	8	9	7	24
Landing Gear	Fixed Skid	10	9	8	27
	Retractable	7	8	9	24

2.2 Material, Body, and Mechanical System Production

2.2.1 Material selection

We used **carbon fiber** and PETG (Polyethylene Terephthalate Glycol).

Table 4 Material Comparison

Criteria	Carbon Fiber Composite (Selected)	Aluminum 7075-T6
Density (g/cm ³)	1.6	2.81
Yield Strength (MPa)	350	275
Corrosion Resistance	High	Moderate
Fatigue Resistance	High (Longer Lifespan)	Moderate
Manufacturing	Moderate (Advanced Molding Needed)	Easier to Machine
Cost	Higher Initial Cost, Long-Term Benefit	Lower Cost
Weight	Lighter (~43% Less Weight)	Heavier

Carbon Fiber is selected due to its superior strength-to-weight ratio.

2.2.2 Frame design feature

The UAV features a compact and full-bodied design with a diagonal length of 493 mm and a height of 50 mm (shown in Figure 2), ensuring a balanced and aerodynamic structure. The base plate, made of carbon fiber, measures 150 × 150 mm. This combination of dimensions and materials ensures durability, stability, and efficient performance during flight.

Table 5 landing gear parameter

Type	Material	Length	Width
3D Printed skid	Carbon Fiber	150mm	290mm

The design features extended curved legs (shown in Figure 3) that provide ample ground clearance, which is suitable for carrying payloads. Due to the symmetrical structure, it also provides stable takeoff and landing.

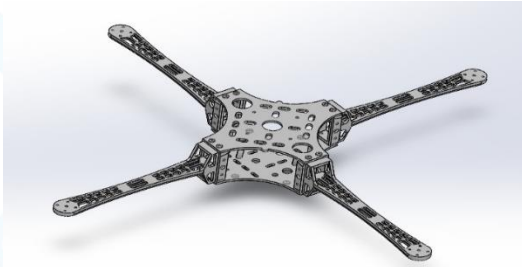


Figure 2 Frame Design

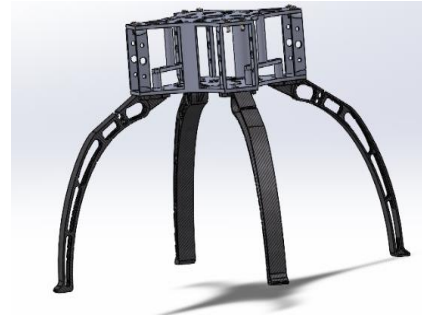


Figure 3 Landing Gear

2.2.3 Mechanical system

Finite Element Analysis (FEA)

To ensure the structural reliability, strength, and optimal performance of the UAV. The von Mises stress analysis (shown in Figure 5) shows the frame experiences a maximum stress of 42 MPa, well below the ultimate tensile strength of carbon fiber (1000 MPa). The mesh detail simulation (shown in Figure 5) used a tetrahedral mesh with an element size of 1.0×10^{-3} . These results confirm the frame's durability and reliability, ensuring it can withstand operational stresses without failure.

The strain analysis

Maximum strain value of 0.0006 is shown in **Figure 4**. This indicates minimal deformation. To calculate the structural reliability, the safety factor (SF) was calculated. Substituting these values $SF=23.8$. A safety factor of 23.8 indicates that the UAV frame can withstand nearly 23.8 times the applied operational stresses before reaching material failure.

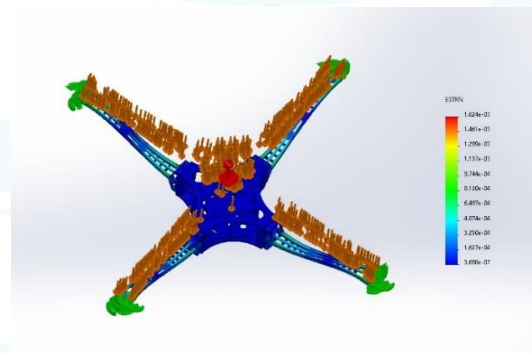


Figure 4 Strain analysis

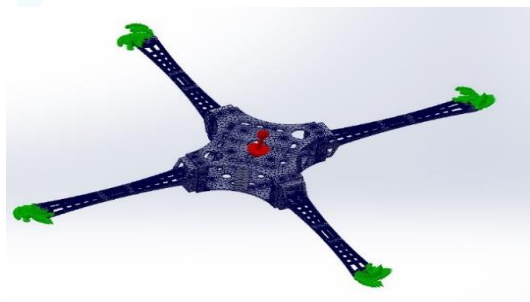


Figure 5 Von Mises stress analysis

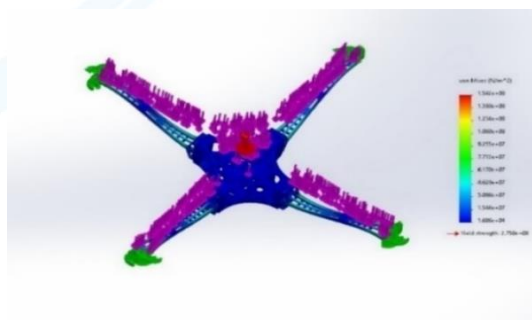


Figure 6 Mesh Detail Analysis

Table 6 Finite Element Analysis (FEA) of carbon fiber frame

Parameter	Carbon Fiber Composite (Selected)
Maximum Stress (MPa)	42 MPa
Yield Strength (MPa)	350 MPa
Safety Factor (SF)	8.33
Maximum Strain	0.0006
Deformation	Minimal

The **Carbon Fiber frame** experiences significantly lower stress and deformation under load

Computational Fluid Dynamics (CFD) Analysis

The CFD analysis was performed using **SolidWorks**. The simulation, conducted with an inlet mass flow of 0.0001 kg/s, atmospheric pressure of 101325 Pa, and a rotating region speed of 1047.19755 rad/s, UAV generated a lift force of 29.43 N. The Lift-to-Drag ratio of 20 and drag coefficient of 0.245 indicate high aerodynamic efficiency and minimal resistance, while the lift coefficient of 4.90 confirms strong lift generation.

Table 7 Computational Fluid Dynamics (CFD) of carbon fiber frame

Parameter	Carbon Fiber Composite (Selected)
Drag Force (N)	1.47 N
Lift Force (N)	Higher (Better Aerodynamics)
Drag Coefficient (CD)	0.245
Airflow Stability	Smooth, Minimal Separation

Carbon Fiber Composite significantly improves aerodynamic efficiency.

Results: These results ensure the UAV's ability to handle dynamic flight conditions, stability, and precise payload delivery, and regain within mission time limits.

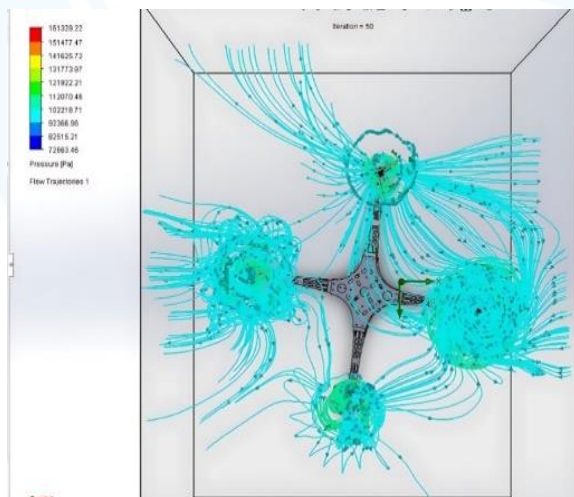


Figure 7 CFD analysis (top view)

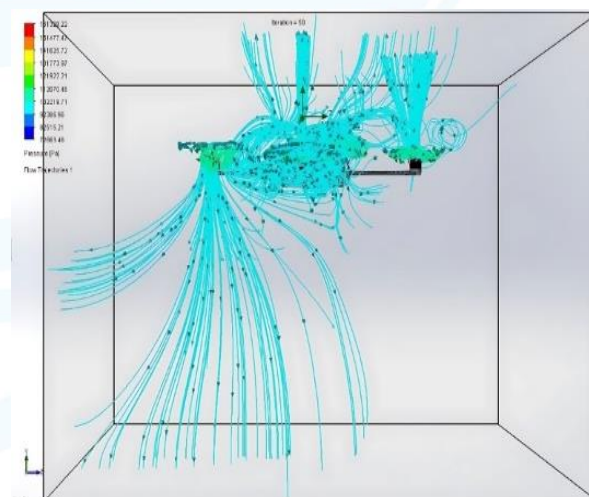


Figure 8 CFD analysis (side view)

$$C_d = (2 \times F_d) / (\rho \times v^2 \times A)$$

$$F_d = (1/2) \times \rho \times v^2 \times C_d \times A$$

2.3 Mission Mechanism System

The mission mechanism system of UAVs is divided into payload management, the **capability** of **autonomous navigation**, and competent mission implementation. The system includes servo-controlled mechanisms, image processing algorithms and manual control options, personalized to meet the competition mission requirements.

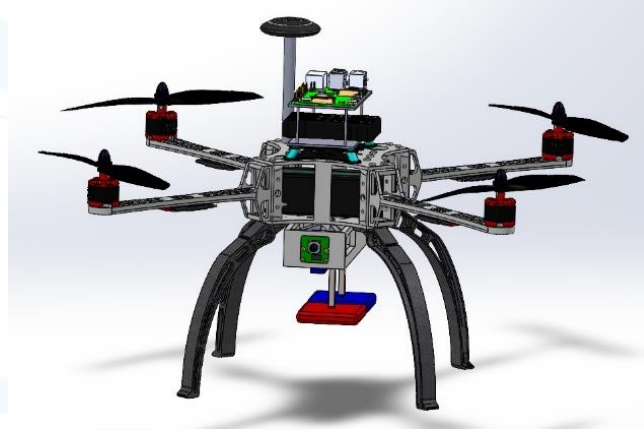


Figure 9 Full Assembly of UAV

Mission Mechanism 1 focuses on the **UAV's autonomous flight capabilities**. We define the flight path, altitude, and velocity using Mission Planner, a ground control software, and upload these parameters to the Pixhawk flight controller. The UAV will autonomously execute the mission, including takeoff and landing. The UAV will make an infinity loop horizontally around reference poles (shown in Figure 10) while carrying two payloads.

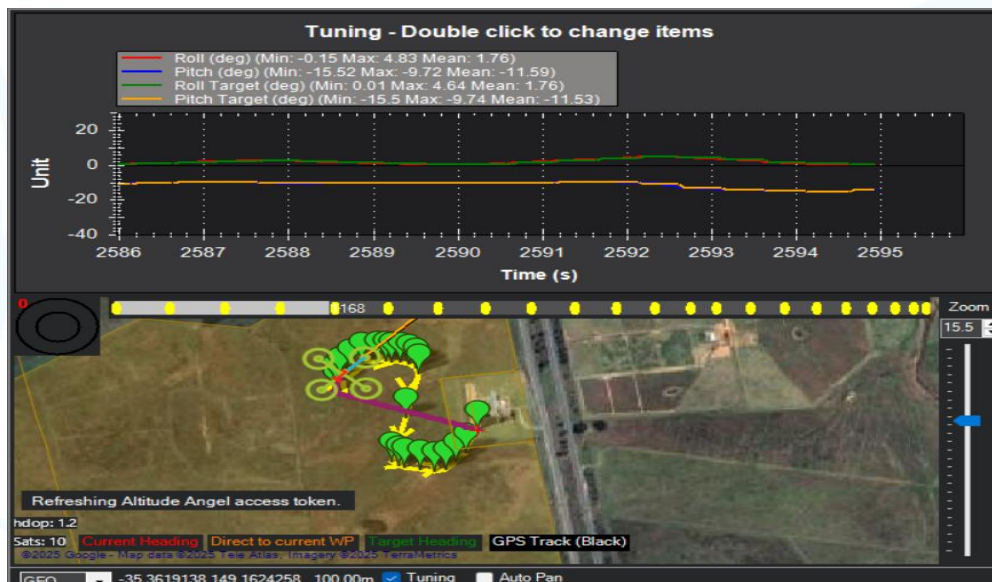


Figure 10 Infinity loop simulation using mission planner

Mission Mechanism 2: The UAV performs advanced image processing and sensor-based targeting to detect the target (shown in Figure 11) and drop payloads using a servo motor rod control mechanism (shown in Figure 13) by confirming the command through Raspberry Pi to the microcontroller when the target is detected, on hexagonal and triangular targets with high accuracy.

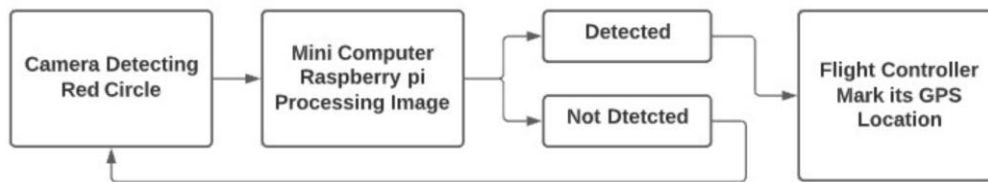


Figure 11 Target locking algorithm



Figure 12 Payload drop to target successfully detected

The release mechanism is designed to handle payloads of up to 500g, ensuring precise drops within a 10 cm radius of the target center.

Mission Mechanism 3: The UAV is armed with a gripper (shown in Figure 14) that can autonomously pick up payloads from the hexagonal target and drop them to the triangular target (shown in Figure 15).

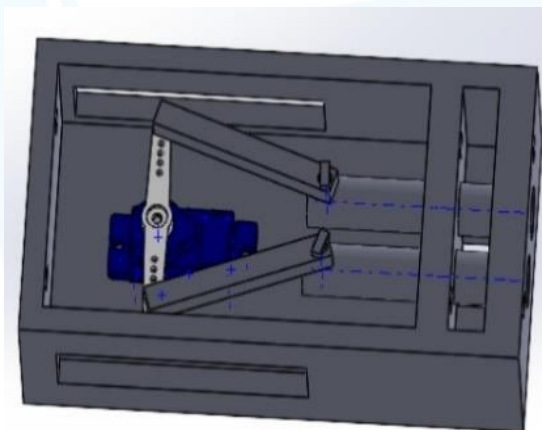


Figure 13 Servo motor road mechanism

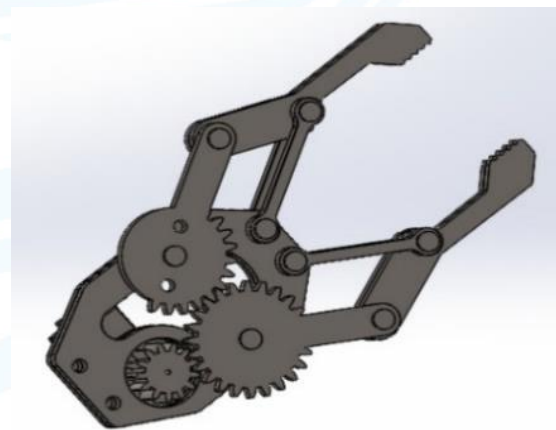


Figure 14 Payload gripper



Figure 15 Autonomously picked up and dropped payload at the target location.

The mechanisms are controlled using a combination of **autonomous systems (Pixhawk flight controller)** and

2.4 Electrical and Electronic Hardware and Power Systems

2.4.1 flight Controllers

The **Pixhawk 2.4.8** is selected as the flight controller due to its integration with multiple sensors and proven stability in professional and hobby UAVs. It is paired with the **NEO-8M GPS module** for accurate position data and the **HMC5883L compass module** for heading information, ensuring stable flight control. These modules are seamlessly integrated with the controller, providing precise navigation and orientation data.

2.4.2 Thrust System

The thrust system consists of four GARTT ML2312 S (960KV) BLDC motors and Simon Emax 40A ESCs. Referring to Table 11, the total loaded weight of the Rotary Wing is $2300 \pm 100g$. The GARTT ML2312 S (960KV) can provide a thrust of 1100g in a single capacity. Four such Motors provide a loading capacity of 4400g, which is safe for operating them at $2500 \pm 100g$. Operating Voltages of Motors are 7.5 to 12V, and a nominal current of 9 -13A. A 3S Lipo Battery with a 35C discharge Rate is best for performance.

Table 8 Specification of BLDC, ESC, and Propeller

BLDC Specification		Speed Controller Specifications	
Model	GARTT ML2312 S (960KV)	Model	BL Heli 40A
Max Current	19.5A	Max Current	40A
Max Thrust	1220g (10" Propeller)	Input Voltage	2S - 6S LIPO

2.4.3 Power Unit

The UAV is powered by a LiPo 3S 5200mAh battery, which provides a continuous discharge rate of 35C and a max burst discharge of 70C. A 5.3V/2.25A power module powers the flight controller, while a 12-5V converter supplies power to the Raspberry Pi and Arduino. The power distribution is managed via a power distribution board, ensuring stable voltage regulation across all components.

Table 9 Specification of LIPO battery

Name	Capacity	Voltage	discharge rate	Max Burst Discharge
LiPo Battery	5200mah	11.1/3S	35C	70C

Table 10 Specification of servo motor

EMAX	Operating Voltage	Operating Current	Operating velocity		Dead band:	Stall torque	
			4.8V	6V		4.8V	6V
ES08MA	4.8V~6.0V	200mA/60°	.12	.1	1-2 uSec	1.5kg.cm	1.8kg.cm

2.4.4 Radio Telemetry

HC-12-based telemetry is used to link the UAV with the **ground station software**. The ground station transceiver was connected via a **CH340G USB-to-Serial** converter to receive or transmit the flight data or instructions to or from the UAV.

2.4.5 Circuit Breaker

120A Miniature Circuit Breaker is used for protection. It is a thermal magnetic MCB-type breaker. The breaker can handle voltages around **12-48V**. It has a fast response time and a manual reset button. It is placed after the battery to avoid drawing the access current.

2.4.6 Payload mechanism

For the payload mechanism, **SG90 9G** Micro Servo Motor is used which is controlled using an **STM32f103** microcontroller. It is a lightweight and fast servo motor, and the STM32F103 is a fast and cheap microcontroller, with a clock of 72MHz. The servo motor will drop the load when Raspberry Pi gives a command to STM32 to drop the payload when the target of a specific color is detected.

2.4.7 Transmitter and Receiver:

We will try to design the RC (shown in Figures 16 and 17) using the NRF2401 2.4GHz transceiver that is used for emergency control. This low-cost, reliable module operates in the 2.4 GHz ISM band, enabling global, unlicensed wireless communication.

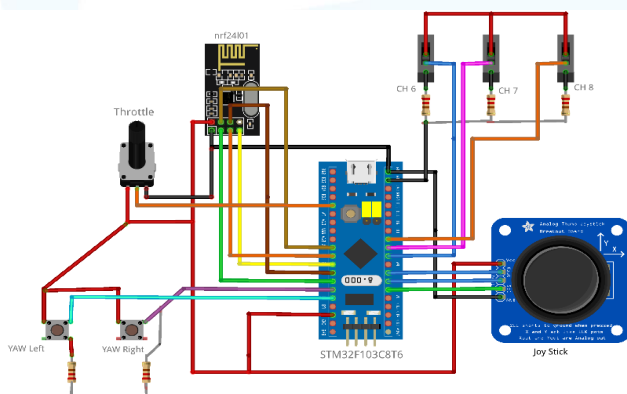


Figure 16 Transmitter Circuit diagram

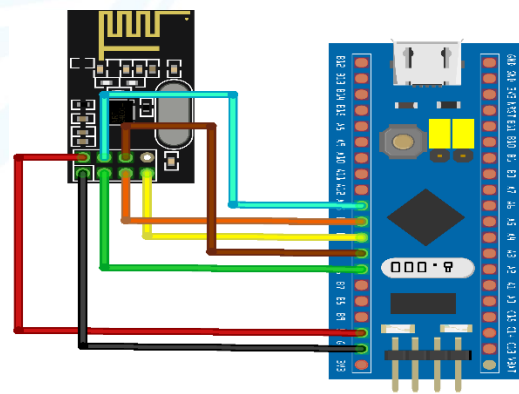


Figure 17 Receiver

2.4.8 Tracking and Target Detection

The target detection system is based on a Raspberry Pi 4 and Raspberry Pi Cam v2. The Raspberry Pi runs an object detection algorithm (YOLO V5 Nano) to detect targets, calculate coordinates, and transmit them to the Pixhawk via UART protocol. The system is powered by a 12-5V buck converter.

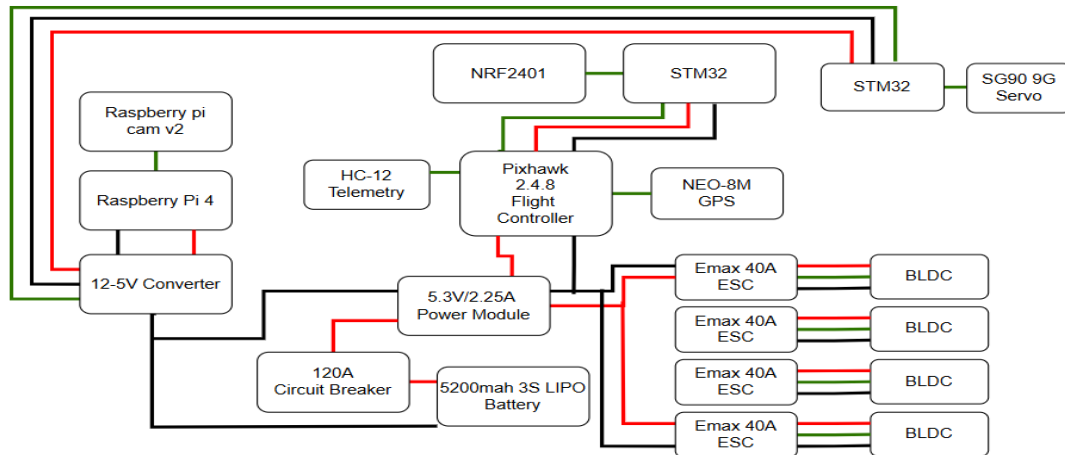


Figure 18 UAV Wiring Diagram

2.5 Software and Control System

2.5.1 Software Architecture & UAV Control

The UAV is first simulated in the gazebo to check the payload drop mechanism. We use ArduPilot and PX4 Autopilot, two open-source flight controlling platforms, to attain UAV stability and navigation. These two platforms are linked with ROS (Robot Operating System) for real-time communication and control of the UAV. The UAV is controlled using Mission Planner, which allows the UAV for mission planning and range finding.

2.5.2 Image Processing Algorithms

For target detection and tracking, our UAV uses deep-learning-based algorithms such as YOLO (You Only Look Once) or Yolo V5 Nano. In addition to this, OpenCV (Open-Source Computer Vision Library) is also used for image processing.

2.5.3 Ground station software & sensor integration

The UAV communicates with the ground station through MAVLink protocol, using Mission Planner and MAVProxy to monitor the flight live data. The **HC-12 telemetry module** ensures reliable communication for data transmission and control. Key sensors include GPS – for positioning and navigation and ultrasonic sensor HC-SR04

2.5.4 Autonomy level

The UAV operates in multiple autonomy levels: Manual Mode – Fully pilot-controlled **Assisted Mode** – Stabilization support (e.g., altitude hold) Semi-Autonomous Mode – Pilot provides high-level commands Fully Autonomous Mode – UAV completes missions independently using AI

2.5.5 Flight control algorithm

The UAV uses PID (Proportional-Integral-Derivative) Controllers for stability and Model Predictive Control (MPC) to minimize flight paths and achieve less flight time. Sliding Mode Control (SMC) ensures UAV stability by rapidly adjusting control inputs in response to disturbances (wind, payload shift).

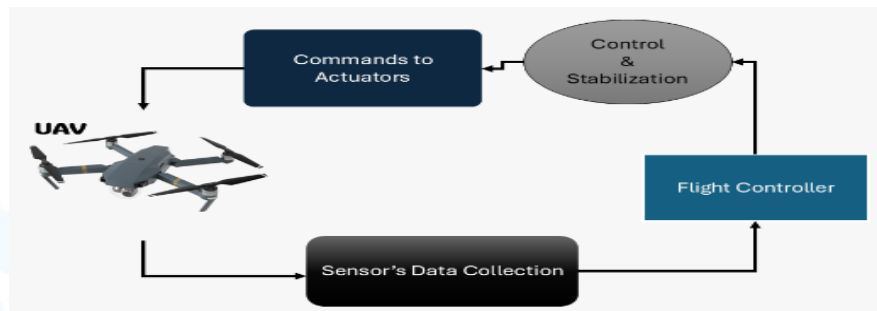


Figure 19 Block diagram of the flight control algorithm

2.6 UAV Performance Parameters

2.6.1 Thrust calculation

if we GARTT ML2312 S (960KV) we will get nearly $1100 \pm 100\text{g}$ thrust per motor

Table 11 Different performance parameters

Throttle Level	Total Thrust (g)	Payload Capacity (g)	Range (km)	Normal Cruise Speed	Thrust per Motor (g)
100% Throttle	4500	350	2.775	14.54 m/s	1125
75% Throttle	3375	350	4.237	12.5 m/s	843.75

2.6.2 Weight Calculation

Table 12 Weight of components

Category	Component Name	Weight (g)
Frame & Legs	Quad-frame + Skid landing legs	400g
BLDC & Propellers	4x 960 kV BLDC, Prop	320g
ESC & Battery	4x Emax 40A ESC & 5200mAh LiPo 3s	550g
Flight Controller	Pixhawk	40g
Vibration Damper	2x Vibration Dampers	25g
GPS & Compass	Neo7m, HMC5883 with mount	40g
Microcontroller	STM-32 F103C8T-6	20g
Raspberry Pi & Camera	Raspberry Pi 4B, Camera	50g
Payload System	Servo motor, gripper, mechanism box, load	350g
Buck Converter	12V to 5V buck converter	20g
Telemetry & Receiver	HC-12	45g
Wire	Wire	100g
Switch	Connectors	60g
Power Components	Power module, Fused Current Breaker	160g
Empty Weight	Without payload	(1830$\pm$100) g
Total Weight	With payload	(2180$\pm$100) g

The maximum loaded weight of the rotary wing is $\leq (2350 \pm 100) \text{ g}$

2.6.3 Power consumption

Maximum Continuous Current Calculation

$$\begin{aligned}\text{Continuous Current} &= (\text{Minimum Capacity} / 1000) \times \text{C-rating} \\ &= (5200\text{mAh} / 1000) \times 35\text{C} = \mathbf{182.4A}\end{aligned}$$

This means our battery can provide approximately a maximum continuous current of 182A.

$$\text{Estimated flight time at maximum continuous current} = 60 / 35 = \mathbf{1.7142 \text{ minutes}}$$

However, our required current when the motor is at 75% thrust, and DF1045 Propellers are used is only 66A. So, we can now calculate the actual flight time based on this discharge rate.

Flight Time Calculation at 75% Thrust

$$\begin{aligned}\text{life} &= (\text{Battery Capacity (Ah)} / \text{Continuous Current (A)}) \times 60 \\ &= (5.2\text{Ah} / 66) \times 60 \\ &= \mathbf{4.72 \text{ minutes}}\end{aligned}$$

At 75% thrust, requiring 66A, the estimated flight time is 4.72 minutes, which is suitable for our application.

2.6.4 Thrust-to-weight ratio

The thrust-to-weight ratio can be calculated as:

$$\begin{aligned}\text{Ratio} &= \frac{\text{Thrust}}{\text{Weight}} = \frac{\text{ma}}{\text{mg}} \\ \text{Ratio} &= \frac{3450}{2300} = \mathbf{1.38}\end{aligned}$$

The thrust-to-weight ratio of 1.38 is good. A high ratio provides a high climb rate and with the thrust to weight ratio of >1.0, UAV can accelerate vertically improving payload lifting capacity.

2.6.5 Weight balancing

The optimized component placement and the calculations ensure a stable Center of Gravity (CoG) for the UAV.

Table 13 Component Placement

Sr. No	Component Name	Weight (g)	X (mm)	Y (mm)	Z (mm)
1	BLDC (Left)	100	-200	0	0
2	BLDC (Right)	100	200	0	0
3	GPS Module	40	0	50	30
4	Flight Controller	40	0	0	0
5	ESC(Left)	80	-150	-50	0
6	ESC(Right)	80	150	-50	0
7	Battery	500	20	0	0
8	Power Module	160	-100	0	0
9	Raspberry Pi & Camera	50	100	-50	20
10	Circuit Breaker	100	-80	0	-25
11	Receiver	30	50	0	0
12	Frame & Legs	400	0	0	0
13	Vibration Damper	25	0	0	0

2.6.6 Payload matter

We selected a rectangular payload due to its compatibility with the UAV's storage mechanism, providing minimum movement and more stability during flight. The rectangular shape provides better weight distribution, enhancing balance and flight stability. It makes gripping and release mechanisms easier, making autonomous delivery more reliable. Additionally, the small-sized design minimizes aerodynamic drag, improving overall efficiency.

2.7 Visual Design Configuration

2.7.1 3D Views

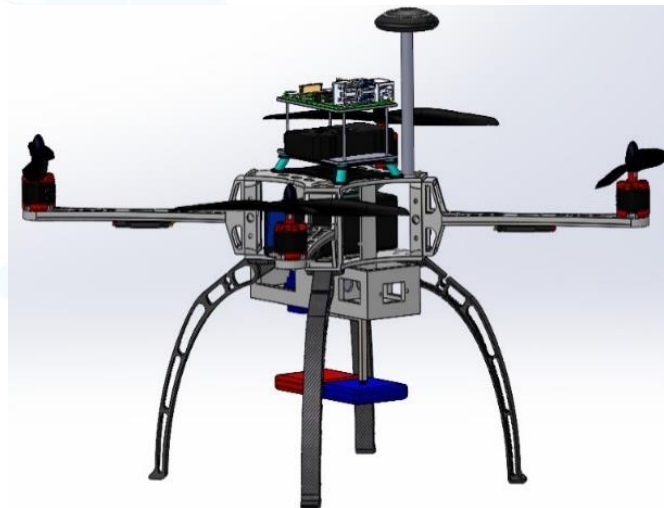
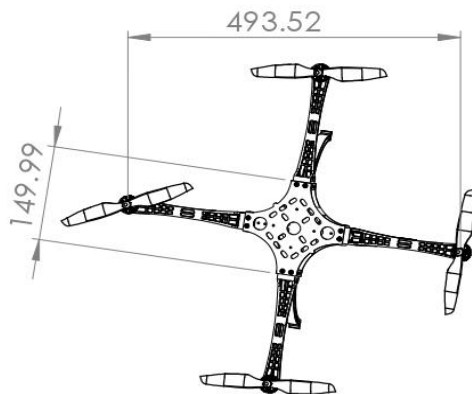


Figure 19 UAV side view



Figure 20 UAV top view

2.7.2 Engineering Drawing



ROTARY WING UNMANNED AERIAL VEHICLE

TEAM: MACH-X

MODEL NAME: UAV

MODEL TYPE : ASSEMBLY

SCALE: 1 : 10

PAGE SIZE: A4

UNIT : mm

3. COST ANALYSIS AND BUDGET TABLE

Table 14 Budget Table

No	Materials	Pieces	Unit Price (TRY)	Total Amount (in TRY)	Reason for Purchasing Material
1	BLDC motor	4	733	2933	Used for propulsion and lift of the UAV.
2	Servo motor	2	74	296	Used in controlling payload mechanism.
3	GPS Module	1	933	933	Required for navigation and positioning.
4	Pixhawk 4 & power module	1	5066	5066	flight controller for autonomous flight.
5	ESC	4	5000	2000	Required to control BLDC.
6	Battery	2	1600	3200	Power source during flight.
7	Raspberry Pi 4B	1	4000	4000	Used for image processing and projectile calculations.
8	Pi Camera(V2)	1	1066	1066	used for target detection.
9	Circuit Breaker	1	600	600	Protect from short circuits and prevent the UAV.
10	Power distribution board	2	33	66	Power distribution to all components.
11	Manufacturing cost of carbon fiber frame (leg + arm + base)	1	4047	4047	Lightweight & durable.
TOTAL PROJECT SUPPORT REQUESTED FROM TUBITAK*				24207 TRY	

